Web Development

An Internship Training and Summer Project Report submitted to

Hello Anjan!
Updated Text

& Engineering

in partial fulfillment for the award of the

degree of

New Text

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

Submitted by

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Under the guidance of

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HR Manager

CHAIBASA ENGINEERING COLLEGE

(Estb. By Govt. of Jharkhand & run by Techno India under PPP)

Approved by AICTE, New Delhi and Accredited by NBA

Affiliated to Jharkhand University of Technology

An ISO 9001:2015 Certified Institute

Bistumpur, Jhinkpani, West Singhbhum -833215

Month, Year

CERTIFICATE

CERTIFICATE

OF INTERNSHIP

THIS CERTIFICATE IS AWARDED TO

Anjan Kumar

For successful completion of internship as a Web Development intern at VaultofCodes.in for a duration of 1 month from 01/06/2025 to 01/07/2025

We found the candidate sincere, hardworking, technically sound and result oriented. We take this opportunity to appreciate his/her work and contributions to thank and wish all the best for your future endeavors

Founder

Google for Education Partner

Manager

AICTE Corporate ID: CORPORATE6511252d7c3271695622445

Format of declaration

DECLARATION

I hereby declare that I have successfully completed my internship at **VaultOfCodes**, located at **Delhi, India**, from **01/06/2025** to **01/07/2025**. During the internship, I actively participated in the assigned tasks and gained practical knowledge and professional experience in the field of **Web Development**. I adhered to the guidelines provided by my mentor/supervisor and maintained a disciplined and professional attitude throughout the internship period. I am grateful for the opportunity to learn and contribute during this internship and I declare that all the information mentioned above is true and correct to the best of my knowledge.

Thank you.

Anjan Kumar

…….

Signature of the Student

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Mrs. Harshita Singh**, my internship supervisor at **VaultOfCodes**, for the invaluable guidance, encouragement, and support provided throughout the internship. His/Her mentorship greatly enhanced my learning experience and allowed me to gain practical insights into **Web Development**.

I am also thankful to the entire team at **VaultOfCode** for their cooperation and for providing a professional environment that enabled me to apply my academic knowledge to real-world scenarios.

I extend my heartfelt thanks to the **Department of Computer Science** and the management of **Chaibasa Engineering College** for organizing and facilitating this internship opportunity. Their continuous support and encouragement played a key role in the successful completion of this internship.

This internship has been a significant part of my academic and professional journey, and I am deeply grateful to all who made it possible.

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CHAPTER I: INTRODUCTION

1.1 Importance of the Internship

In today's digital landscape, web development plays a pivotal role in shaping how users interact with services, businesses, and content online. The demand for skilled front-end developers has surged, making hands-on experience essential for computer science students. This internship provided a practical platform to learn web technologies beyond theoretical classroom learning and bridge the gap between academia and industry.

1.2 Objectives of the Training

The primary objectives of this internship were:

- To gain hands-on experience in designing and developing real-world websites.
- To understand the foundational technologies of front-end development: HTML, CSS, and JavaScript.
- To work on responsive design principles for different screen sizes.
- To explore project management using Git and GitHub.
- To develop, test, and deploy live web applications.

1.3 Scope of the Training

This internship focused on front-end web development, including:

- Designing static and interactive web pages.
- Creating user interfaces using HTML and CSS.
- Implementing dynamic behavior with JavaScript.
- Managing code versioning and deploying web apps using GitHub.
- Building complete mini-projects such as a Portfolio Website, Internship Promotion Page, and Smart To-Do List Application.

The training not only enhanced technical skills but also improved problem-solving, UI/UX awareness, and understanding of real-world development workflows.

1.4 Period of the Training

The internship was conducted over a period of **30 days**, starting from **10th June 2025** and concluding on **10th July 2025**. The entire internship was virtual and hosted by **Vault of Codes**, allowing flexibility and consistent mentorship throughout the duration.

CHAPTER II: PROJECT MOTIVATION AND BENEFITS

This chapter provides insights into the motivation behind the project work undertaken during the internship at Vault of Codes, the benefits derived from completing it, and the various tools, software, and technologies used during the development of the assigned projects. The chapter also highlights how the project aligns with real-world applications and future professional endeavors.

2.1 MOTIVATION FOR THE PROJECT WORK

The primary motivation behind undertaking web development projects during this internship stemmed from the desire to gain real-world exposure to client-side programming, user experience design, and responsive interface development. In the digital age, web development is a crucial skill set, enabling developers to build accessible and user-centric applications across multiple devices and platforms.

Working on individual projects such as a personal portfolio website, an internship promotion landing page, and a smart to-do list app provided a structured opportunity to:

- Translate academic theory into practice.
- Solve real-time user interaction problems.
- Build visually appealing, functional, and responsive web interfaces.
- Learn modern development tools and workflows.

The internship encouraged the development of independent problem-solving skills, design thinking, and the ability to follow a development cycle from planning to deployment.

2.2 BENEFITS OF THE PROJECT WORK

Engaging in practical project work helped consolidate theoretical learning into meaningful applications. Some of the notable benefits include:

- **Skill Enhancement:** Improved proficiency in HTML5, CSS3, JavaScript, and browser debugging tools.
- **Version Control Proficiency:** Learned effective use of Git and GitHub for project collaboration and code management.
- **Deployment Experience:** Gained experience in deploying live projects using GitHub Pages.
- **UI/UX Awareness:** Developed an understanding of design principles, layout hierarchy, accessibility, and responsive behavior.
- **Productivity:** Built a smart to-do app that included countdown timers, priority settings, and task tracking — useful for time and task management.
- **Portfolio Development:** Published all completed projects to GitHub and used them as part of a professional portfolio.

These benefits not only improved technical confidence but also laid a foundation for future internships, freelance work, and job placements.

2.3 SOFTWARE, LANGUAGES, FRAMEWORKS, AND TOOLS USED

The table below summarizes the primary technologies and tools used in the project work:

Category	Technology / Tool	Purpose / Use Case
Programming Language	Programming Language	Structure and layout of web pages
Programming Language	CSS3	Styling and layout, responsive design
Programming Language	JavaScript	Interactivity, dynamic elements, countdown timers
API/Browser Storage	Local Storage API	Storing tasks and settings locally on the user's browser
Code Editor	Visual Studio Code (VS Code)	Writing, testing, and editing code
Version Control	Git and GitHub	Version tracking, collaboration, and code backup
Deployment Platform	GitHub Pages	Hosting live websites directly from GitHub repositories
UI Libraries	Google Fonts / CSS Grid	Fonts and layout management
Development Tools	Browser Developer Tools	Debugging and performance testing

(Table. 2.1)

2.4 PROJECT MODULES AND COMPONENTS

The main projects included the following components and modules, each crafted to simulate real-world web development workflows and test a variety of technical competencies.

❖ **Portfolio Website**

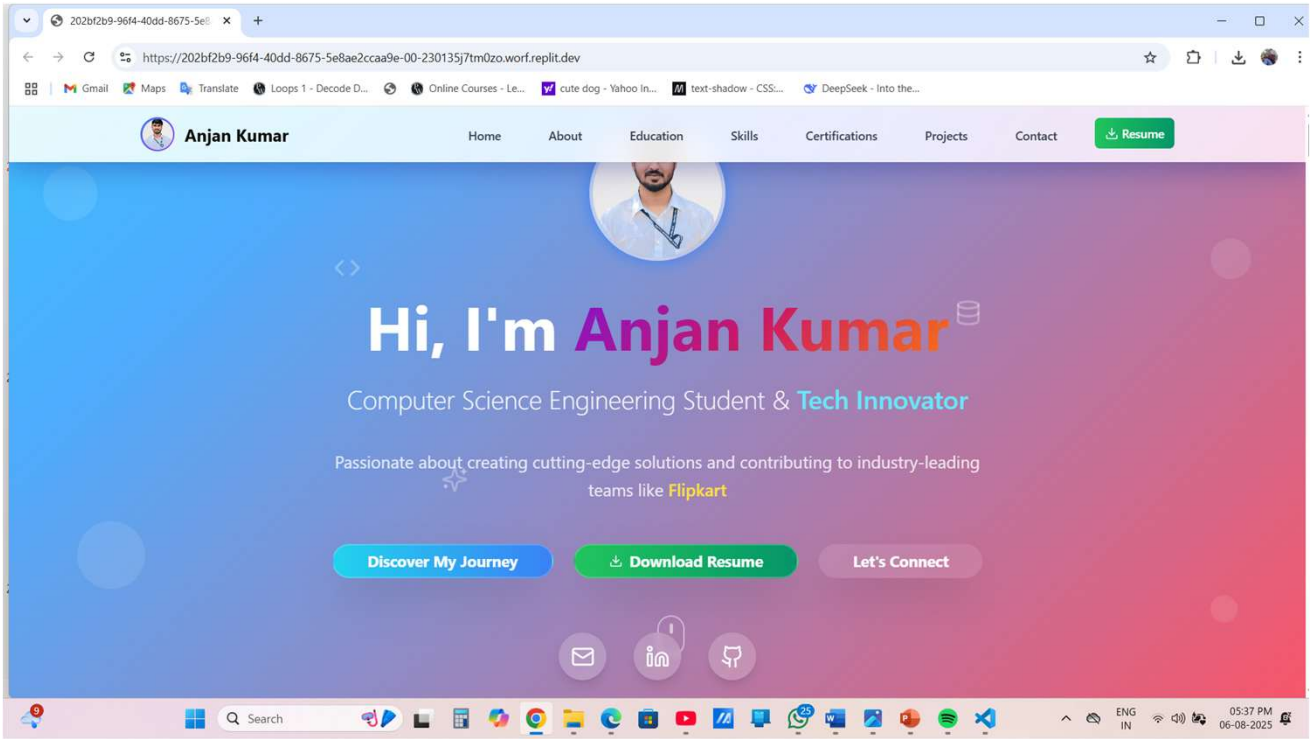
Home, About, Projects, and Contact sections.

Smooth scroll and internal linking for seamless navigation.

Embedded GitHub and LinkedIn profile buttons for professional integration.

Responsive layout ensuring visibility across desktop and mobile devices.

Attached screenshot of Portfolio Website:



(Fig. 2.1)

❖ Internship Promotion Landing Page

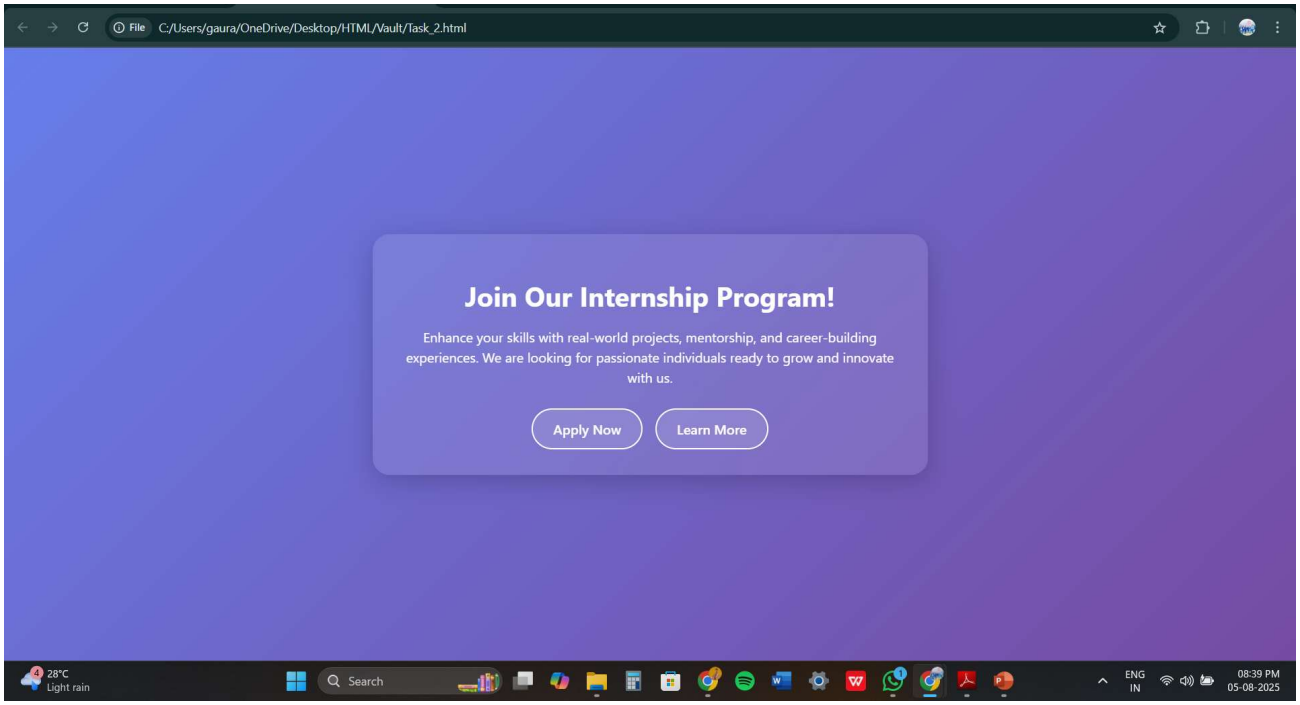
Eye-catching call-to-action sections like "Apply Now" and "Learn More".

Use of gradient backgrounds and advanced CSS effects for modern aesthetics.

Mobile-friendly layout built with CSS Flexbox and Grid.

Designed with conversion optimization principles to simulate real promotional campaigns.

Attached screenshot of Landing Page:

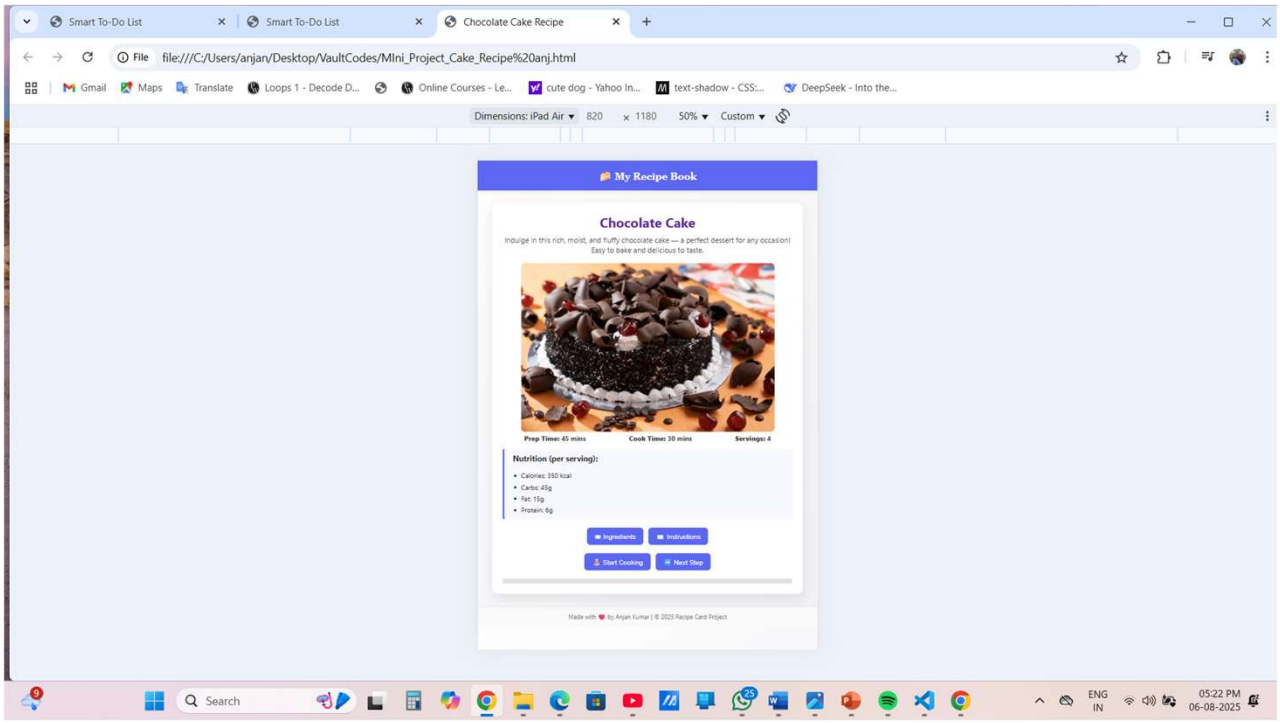


(Fig. 2.2)

❖ Mini Project: "My Recipe Book"

- Designed as a lightweight, user-friendly note-taking application.
- Enables users to create, edit, pin, and delete personal notes on the go.
- Implemented responsive layout using **Flexbox and Media Queries** to ensure full functionality across devices.
- Leveraged **JavaScript LocalStorage API** to store user notes persistently without requiring backend/database.
- Included real-time character counter and dynamic color tags for easy note categorization.
- Focused on clean UI with card-style note blocks and intuitive interaction buttons.
- Designed as a productivity-enhancing tool, suitable for students, freelancers, and professionals

Attached screenshot of My Recipe Book :



(Fig. 2.3)

❖ Smart To-Do List Web App

Task input form with real-time validation to ensure clean data input.

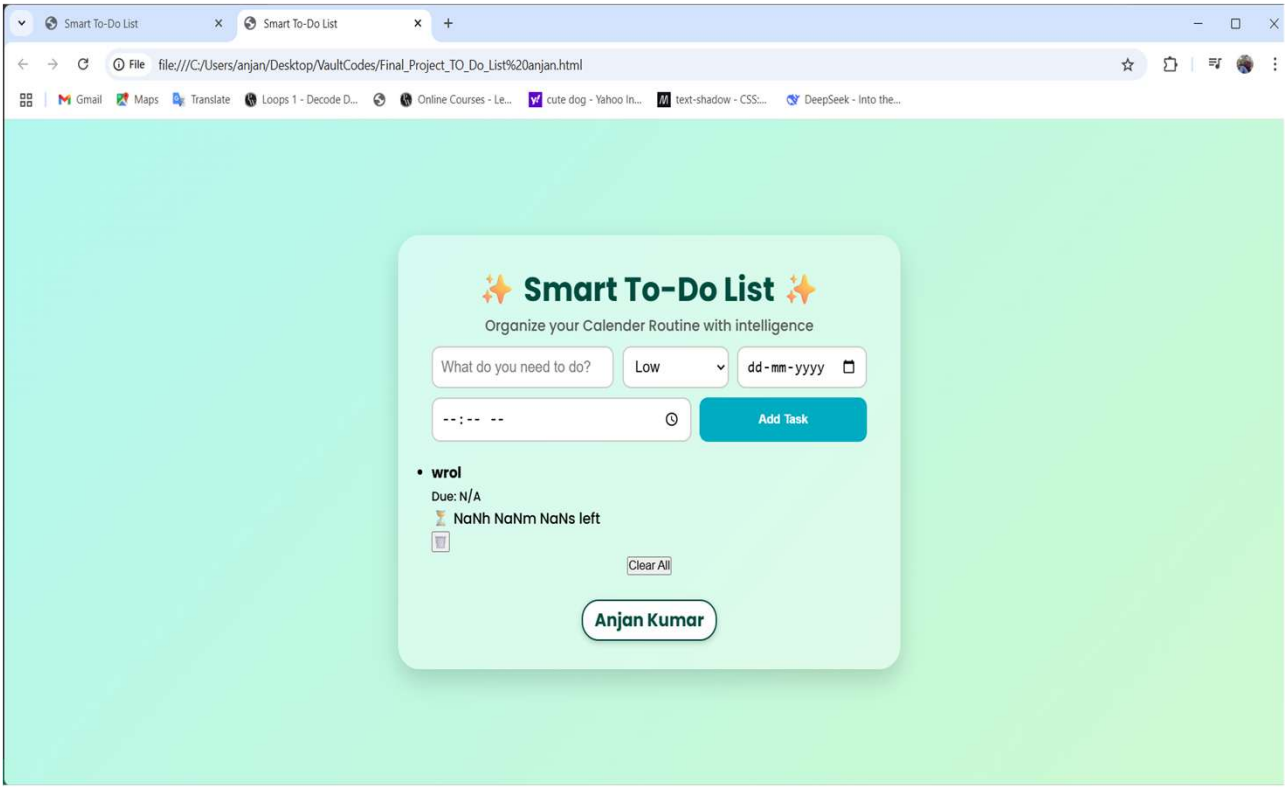
Priority dropdowns and due date pickers for personalized scheduling.

Countdown timer using JavaScript to show urgency and deadlines visually.

Integration of LocalStorage API to persist user data without a backend.

Modular code structure enabling future enhancements like notifications or dark mode.

Attached screenshot of Smart To-Do List Web App:



(Fig. 2.4)

Each of these modules reflects a strong emphasis on responsive design, efficient user experience, and technical modularity. By working independently on these functional applications, the internship offered a real glimpse into the lifecycle of front-end development: from ideation and layout planning to coding, debugging, deployment, and UI refinement.

The combination of these tools and the motivation to develop meaningful web applications contributed to a highly beneficial internship experience. The hands-on nature of the project work enabled deeper learning, better retention, and greater confidence in taking on real-world software development tasks.

CHAPTER III: RESULT AND ANALYSIS OF PROJECT WORK

This chapter presents a detailed analysis of the outcomes of the web development internship projects undertaken at Vault of Codes. It highlights the development phases, methodologies used, visual diagrams, functional workflows, and real-world applications of the completed projects. Each stage of development is assessed for its functionality, logic flow, and contribution to the learning experience.

3.1 PROJECT DEVELOPMENT PHASES

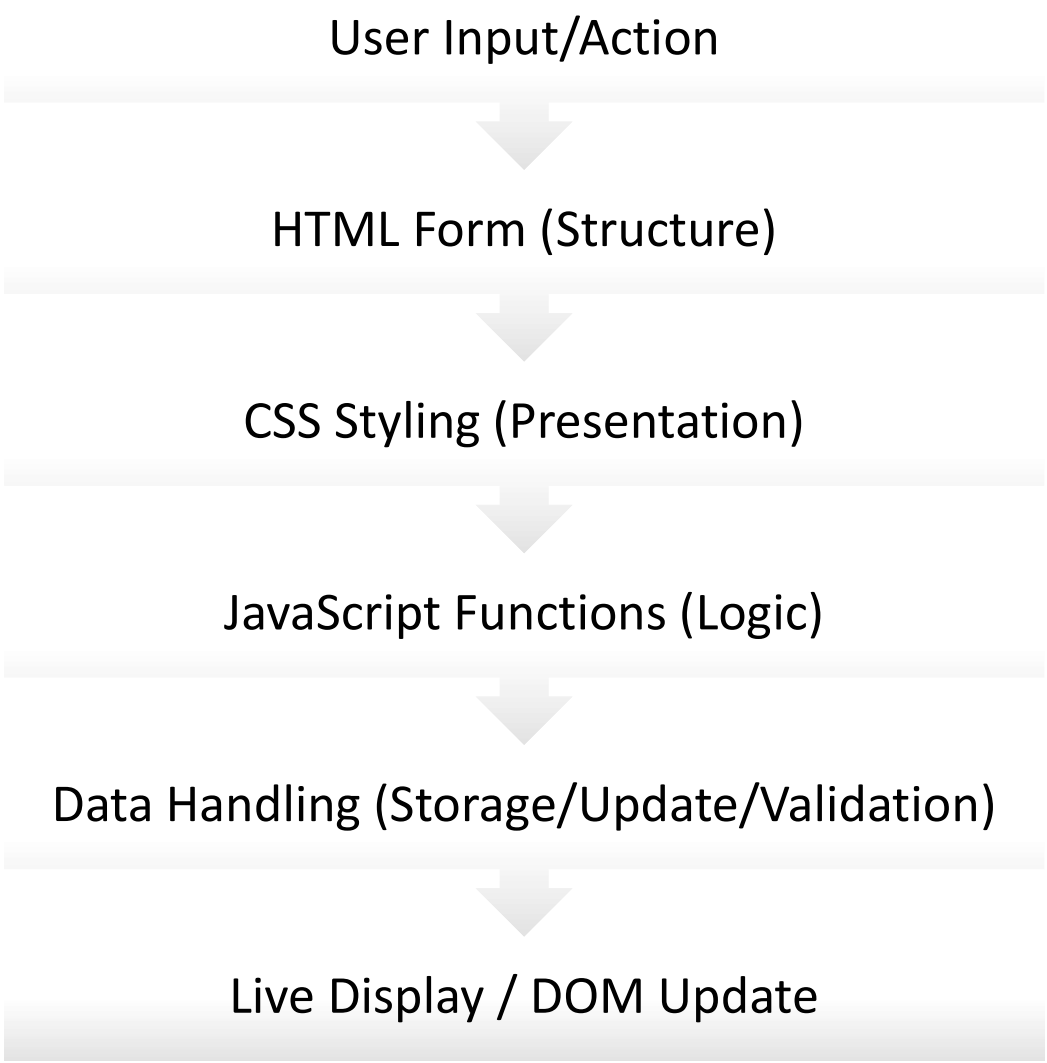
The entire project development lifecycle was structured into the following phases:

1. **Requirement Analysis:** Understanding the scope of each mini-project.
2. **Planning and Design:** Preparing layout sketches, wireframes, and choosing color schemes.
3. **Development:** Writing modular, scalable code using HTML, CSS, and JavaScript.
4. **Testing and Debugging:** Using browser tools and test inputs to ensure cross-browser compatibility.
5. **Deployment:** Hosting projects live on GitHub Pages.
6. **Documentation:** Writing README files, report content, and code comments

3.2 FLOWCHART AND WORKFLOW DIAGRAMS

Each project followed a simplified front-end development workflow:

General Workflow Diagram:

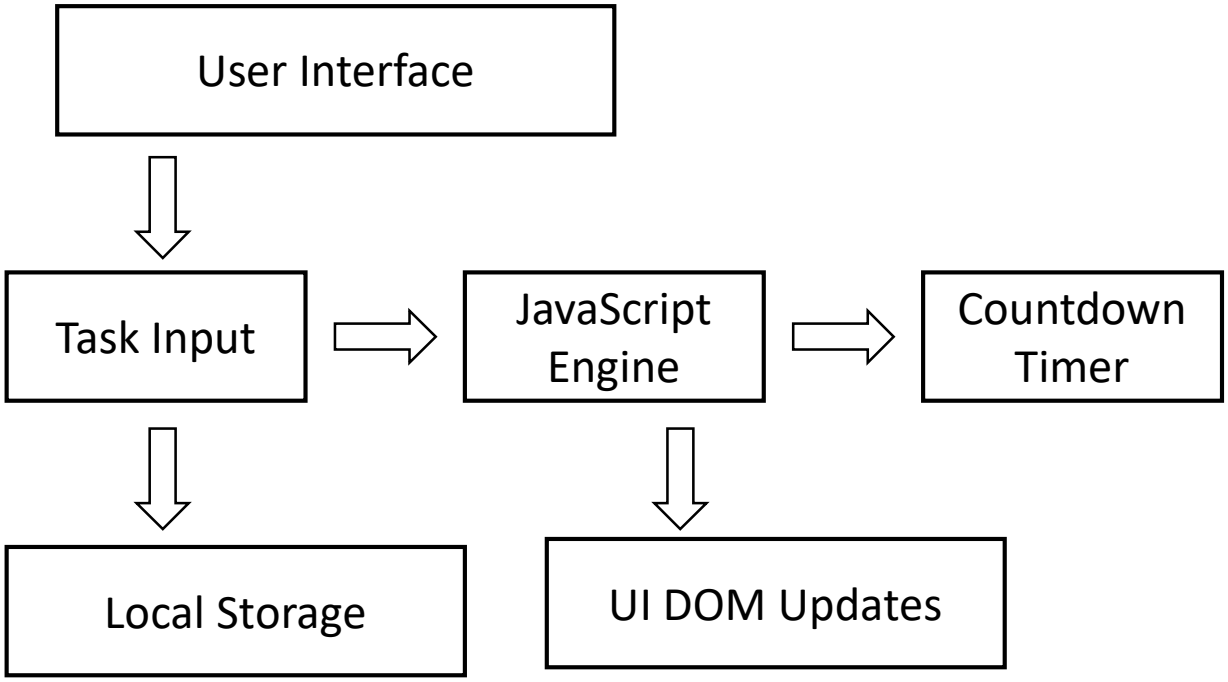


(Fig. 3.1)

This structure ensures a clear separation of concerns: content, presentation, and behavior.

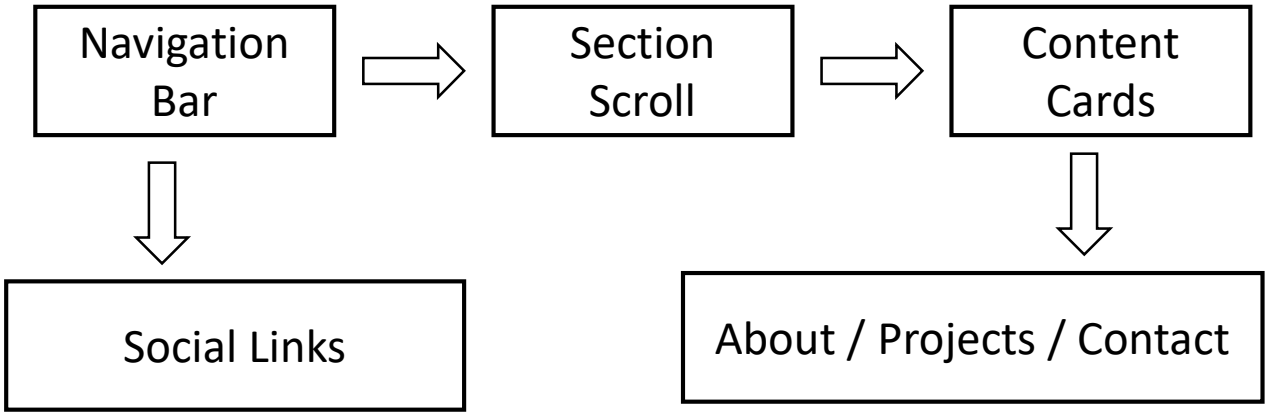
3.3 FUNCTIONAL BLOCK DIAGRAMS

Smart To-Do List Web App – Block Diagram:



(Fig. 3.2)

Portfolio Website – Block View:



(Fig. 3.3)

3.4 PROJECT FUNCTIONALITY AND OUTPUTS

All projects were successfully executed and deployed. The following are the key outputs:

- **Portfolio Website:** Live, responsive portfolio site with interactive links and structured layout.
- **Landing Page:** A modern, visually appealing promotional page encouraging internship registrations.
- **To-Do App:** Fully functional task manager with countdown timers, task prioritization, and persistent storage.

These outputs demonstrate the intern's ability to work independently and produce tangible, user-facing products.

3.5 REAL-WORLD APPLICATIONS

The projects completed during the internship have practical relevance in real-world environments:

- **Portfolio Website:** Can be used for job applications, personal branding, and online visibility.
- **Landing Page:** Template for real product or service promotion by startups, NGOs, or freelance developers.
- **Smart To-Do App:** Usable productivity tool for students and professionals to manage tasks and deadlines.

These applications are not only relevant academically but can be directly used, enhanced, or integrated into larger web-based platforms.

3.6 ALGORITHM / LOGIC IMPLEMENTATION

Each JavaScript-based interaction (e.g., countdown timer, task validation) followed logical algorithmic structures. Example logic:

Countdown Timer Logic in To-Do App:

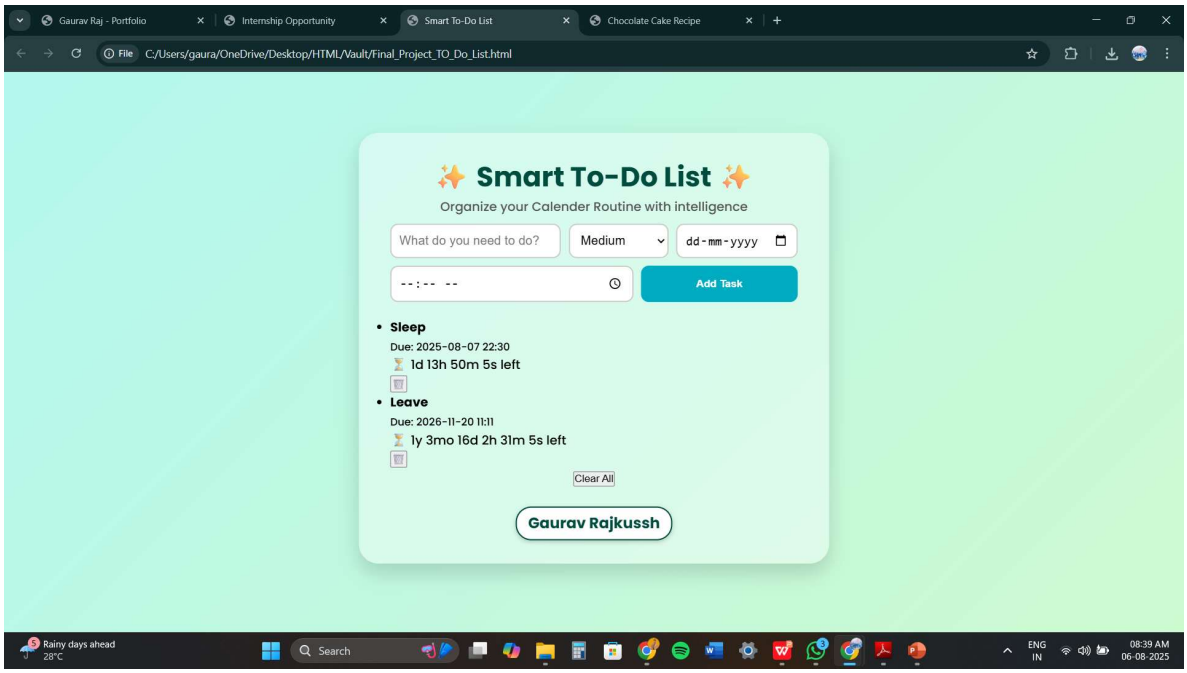
Retrieve current time.

Subtract from task deadline.

Convert result into hours/min/sec.

Update UI every 1 second using

Clear interval when time reaches zero.



(Fig. 3.4)

This structured approach improved algorithmic thinking, debugging skills, and real-time interaction handling.

The results clearly reflect that the projects were not only technically accurate but also thoughtfully designed, functionally robust, and aesthetically clean. This comprehensive development experience has built a solid foundation for future software development roles.

CHAPTER IV: CONCLUSION

The internship at Vault of Codes provided a holistic learning experience that extended beyond theoretical knowledge and into practical, real-world application. By participating in web development projects, the internship effectively bridged the gap between academic learning and industry-oriented practices. Each department played a critical role in shaping the overall experience, and the knowledge gained has been both diverse and impactful.

4.1 OBSERVATIONS FROM DIFFERENT DEPARTMENTS

4.1.1 Technical Development Department

The technical team provided clear guidelines for all projects, including detailed instructions, code templates, and learning modules. Their review system ensured that the code was functional, scalable, and followed best practices. This department served as the backbone of the development process, simulating a professional tech team environment.

4.1.2 Curriculum and Content Team

The structured curriculum, divided into weekly modules, helped maintain a steady learning pace. The documentation, including PDFs and online tutorials, was comprehensive and beginner-friendly. This department ensured that the learning journey was logical and progressive.

4.1.3 Mentorship and Evaluation Team

Mentors were approachable and helpful in resolving technical doubts. Their personalized feedback helped identify mistakes, improve UI designs, and optimize code logic. The evaluation system was fair, timely, and insightful.

4.1.4 Support and Administrative Team

This team ensured seamless communication and coordination. From onboarding to weekly task reminders, the support team maintained consistency and organization throughout the internship. Their proactive approach helped avoid any confusion or delays.

4.1.5 HR and Coordination Team

The HR department played an essential role in aligning interns with project tracks, managing intern records, and issuing certificates. Their presence added a professional structure to the internship, making it feel organized and goal-driven.

4.1.6 Marketing and Outreach Team

Though interaction with this team was minimal, their contribution was evident in the visibility of intern achievements and promotions on digital platforms. It also highlighted how web development contributes to business communication and branding.

4.2 EXTENT OF FULFILLMENT OF TRAINING OBJECTIVES

The internship program successfully fulfilled the training objectives in the following ways:

- **Skill Development:** Gained hands-on experience with HTML, CSS, JavaScript, GitHub, and modern web development tools.
- **Practical Exposure:** Built and deployed real projects like a portfolio site, landing page, and task management app.
- **Professional Workflow:** Learned about the development life cycle from planning and coding to deployment and feedback.
- **Communication & Collaboration:** Regular updates, feedback loops, and structured tasks simulated real company operations.
- **Portfolio & Resume Enhancement:** Projects completed and deployed were added to GitHub and portfolio repositories, providing tangible proof of skills.

The learning experience was highly practical, industry-relevant, and personally enriching. It offered a simulated work environment that not only built technical competence but also instilled professionalism, time management, and independent learning.

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LIST OF ABBREVIATIONS

ABBREVIATED FORM	EXPANDED FORM
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UI	User Interface
UX	User Experience
API	Application Programming Interface
DOM	Document Object Model
HR	Human Resources
VS Code	Visual Studio Code
GIT	Global Information Tracker (commonly known as Git)
URL	Uniform Resource Locator
CRUD	Create, Read, Update, Delete
PDF	Portable Document Format
JSON	JavaScript Object Notation

6.1 Internship progress log book for student

Name of Student:	Anjan kumar
College ID:	22011440013
Organization name and address:	VaultOfCodes
Title of internship (Project/Task/Work):	Web Development
Internship Period and Duration:	1 month
Internship Location:	Online
Name of Supervisor(s)/ Manager(s)/Concerned person(s):	Varshita Singh HR Manager

Day	Date	In Time	Out Time	Observation/Work/Task Performed	Remark
1					
2					
3					
5					
6					
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30					

Signature of Supervisor/ Manager/Concerned person

6.2 Internship feedback form for student

Name of Student:	Anjan kumar
College ID:	22011440013
Organization name and address:	VaultOfCodes, Delhi, India
Title of internship (Project/Task/Work):	Web Dewelopment
Internship Period and Duration:	1 month
Internship Location:	Online

Sl. No.	Questions	Response your experience (Points)					Remarks/ Comment
		Strongly Agree (5)	Agree (4)	No Opinion (3)	Disagree (2)	Strongly Disagree (1)	
1	Given me the opportunity to explore a career field?	✓					Gained practical exposure in web development
2	Allowed me to apply classroom theory to practice?	✓					Implemented HTML, CSS, JS concepts in real projects
3	Helped me develop my decision-making and problem-solving skills?		✓				Solved layout bugs and code logic independently
4	Expanded my knowledge about the work world prior to permanent employment?	✓					Understood client-oriented project building
5	Helped me develop my written and oral communication skills?		✓				Learned documentation and writing clean code comments
6	Provided a chance to use leadership skills (influence others, develop ideas with others, stimulate decision-making and action)?		✓				Managed task planning and project structure
7	Expanded my sensitivity to the ethical implications of the work involved?		✓				Learned about data privacy and responsible coding
8	Made it possible for me to be more confident in new situations?	✓					Boosted confidence in presenting web projects
9	Given me a chance to improve my interpersonal skills?		✓				Communicated regularly with mentors

10	Helped me learn to handle responsibility and use my time wisely?	✓					Followed task schedules and deadlines
11	Helped me discover new aspects of myself that I didn't know existed before?		✓				Realized creativity in UI/UX design
12	Helped me develop new interests and abilities?	✓					Developed interest in frontend development
13	Helped me clarify my career goals?	✓					Now aiming for a full-stack development role
14	Provided me with contacts which may lead to future employment?		✓				Connected with tech mentors and coordinators
15	Allowed me to acquire information or use equipment not available at my Institute		✓				Used GitHub Pages, VS Code, and modern frameworks

Considering your overall experience, how would you rate this internship?

This internship gave me immense practical experience in web development. It bridged the gap between theoretical learning and real-world implementation. I could confidently build, deploy, and present full-fledged web projects, and learned how tech teams collaborate in a virtual environment.

Give your suggestions/Comments: (If any)

The internship was well-organized and smooth. It would be helpful if group-based or collaborative tasks were added to simulate team development.

Date:

Signature of the Student